

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS1

SUBJECT: Chemistry

TOPIC: The Periodic Table (Basics)

Introduction

- The periodic table is an arrangement of all known elements in order of increasing atomic number, with elements of similar properties in the same vertical column (group). It was first devised systematically by Mendeleev, who arranged elements by atomic mass and predicted undiscovered elements.

Structure of the table

- **Periods** — horizontal rows: elements across a period show a gradual change from metallic to non-metallic properties (period 1 has 2 elements, period 2 has 8, period 3 has 8).
- **Groups** — vertical columns 1–8 (0): elements in the same group have the same number of valence electrons and similar chemical properties.
- Group 1 — alkali metals (Li, Na, K): soft, reactive, form +1 ions. Group 2 — alkaline-earth metals. Group 7 — halogens (F, Cl, Br, I): reactive non-metals, form –1 ions. Group 0 — noble gases (He, Ne, Ar): very unreactive.

Periodic law

- The modern periodic law states that the properties of elements are periodic functions of their atomic numbers — hence a repeating pattern of properties down the periods and groups.

CLASS EVALUATION

1. Define the periodic table.
2. Differentiate between a period and a group.
3. State two characteristics of Group 1 elements.
4. Why are the noble gases unreactive?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Chemical Bonding

Introduction

- Atoms combine to form molecules and compounds because this gives them a stable outer shell of electrons (usually 8 — the octet rule). The forces holding atoms together are chemical bonds.

Types of bonds

- Ionic (electrovalent) bonding** — transfer of electrons from a metal to a non-metal: Na gives one electron to Cl, forming Na⁺ and Cl⁻ which attract (NaCl); ionic compounds are crystalline, have high melting points and conduct electricity when molten or in solution.
- Covalent bonding** — sharing of electrons between non-metals: H₂, O₂, CH₄, CO₂; covalent substances may be gases, liquids or solids with low melting points and do not conduct electricity (except graphite).
- Metallic bonding** — a lattice of positive ions in a 'sea' of delocalised electrons: explains the malleability, ductility and conductivity of metals.
- Dative (coordinate) bond** — a shared pair from one atom only (in ammonium ion NH₄⁺).

Properties related to bonding

- Ionic: high melting points, soluble in water. Simple covalent: low melting points, poor conductivity. Giant covalent (diamond, graphite): very high melting points; graphite conducts because of free electrons.

CLASS EVALUATION

1. Explain the octet rule.
2. Distinguish between ionic and covalent bonding with examples.
3. Why does sodium chloride conduct electricity when molten but not when solid?
4. What is a coordinate bond? Give one example.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Chemical Formulae & Equations

Introduction

- A chemical formula shows the kind and number of atoms in a substance (H_2O — 2 hydrogen atoms and 1 oxygen atom). A chemical equation is a shorthand statement of a reaction, showing reactants $\rightarrow$ products with the same number of each atom on both sides.

Writing formulae

- Use the valency of the elements or radicals: sodium (Na, valency 1) and chlorine (Cl, valency 1) $\rightarrow \text{NaCl}$; calcium (2) and oxygen (2) $\rightarrow \text{CaO}$; aluminium (3) and oxygen (2) $\rightarrow \text{Al}_2\text{O}_3$. Common radicals: OH^- , NO_3^- , SO_4^{2-} , CO_3^{2-} , NH_4^+ . The valencies (or charges) must balance.

Balancing equations

- Balance by adjusting coefficients, never subscripts: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ becomes $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. State symbols may be added: (s) solid, (l) liquid, (g) gas, (aq) aqueous. Example: $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$ (g).

CLASS EVALUATION

- Write the formula of: magnesium oxide, calcium chloride, aluminium sulphate.
- Balance: $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$.
- What do the state symbols mean?
- Why must an equation be balanced?

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS1

SUBJECT: Chemistry

TOPIC: The Mole Concept

Introduction

- The mole is the amount of substance containing Avogadro's number (6.02×10^{23}) of particles (atoms, molecules, ions). It links the mass of a substance to the number of particles.

Key relationships

- Molar mass (g/mol) = the mass of one mole; relative atomic mass or molecular mass expressed in grams.
- Number of moles $n = \text{mass} \div \text{molar mass}$.
- Mass = $n \times \text{molar mass}$; number of particles = $n \times 6.02 \times 10^{23}$.
- Examples: 1 mole of water (H_2O , 18 g) = 6.02×10^{23} molecules; 2 moles of CO_2 = 88 g.

Molar volume of gases

- At room temperature (r.t.p.) 1 mole of any gas occupies about 24 dm^3 ; at s.t.p. (0°C , 1 atm) it occupies 22.4 dm^3 . This allows conversions from volume to moles and vice-versa for gases.

CLASS EVALUATION

1. Define the mole.
2. How many moles are in 44 g of CO_2 (C = 12, O = 16)?
3. What volume does 0.5 mole of hydrogen occupy at r.t.p. ($24 \text{ dm}^3/\text{mol}$)?
4. State Avogadro's number.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Mole Calculations & Stoichiometry

Introduction

- Stoichiometry is the calculation of the quantities of substances involved in chemical reactions, using balanced equations and the mole concept. It answers 'how much product will form from this mass of reactant?'.

The method

- 1. Write and balance the equation. 2. Convert the given mass (or volume) into moles. 3. Use the mole ratio from the equation to find the moles of the required substance. 4. Convert back to mass or volume.

Worked example

- How many grams of oxygen are produced when 24.5 g of potassium chlorate decomposes?
 $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$. Moles $\text{KClO}_3 = 24.5 \div 122.5 = 0.2$; ratio 2:3 $\rightarrow$ moles $\text{O}_2 = 0.3$;
mass $\text{O}_2 = 0.3 \times 32 = 9.6 \text{ g}$.

Limiting reactant & percentage yield

- The limiting reactant is the one used up first; it determines the amount of product.
Percentage yield = $\text{actual mass} \div \text{theoretical mass} \times 100$. Excess reactant remains unreacted.

CLASS EVALUATION

- Calculate the mass of CaO from 50 g of CaCO_3 ($\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$).
- Define the limiting reactant.
- What is percentage yield?
- Convert 4.4 g of CO_2 to moles (C = 12, O = 16).

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Valency, Radicals & Writing Formulae (Revision)

Introduction

- Valency is the combining power of an element or radical — the number of hydrogen atoms (or their equivalent) it can combine with or displace. It follows from the number of electrons in the outer shell.

Valencies of common elements

- Group 1: +1 (Na, K); Group 2: +2 (Mg, Ca); Group 3: +3 (Al); Group 6: -2 (O, S); Group 7: -1 (Cl, Br); nitrogen can be -3, +3, +5; transition metals have variable valency ($\text{Fe}^{2+}/\text{Fe}^{3+}$, $\text{Cu}^{+}/\text{Cu}^{2+}$).

Common radicals

- Hydroxide OH^{-} , nitrate NO_3^{-} , hydrogen carbonate HCO_3^{-} , carbonate CO_3^{2-} , sulphate SO_4^{2-} , sulphite SO_3^{2-} , phosphate PO_4^{3-} , ammonium NH_4^{+} . In a formula the total positive valency equals the total negative valency: $\text{Al}^{3+} + \text{SO}_4^{2-} \rightarrow \text{Al}_2(\text{SO}_4)_3$ (brackets for a repeated radical).

CLASS EVALUATION

- Define valency.
- Write the formula of: iron(III) oxide, ammonium sulphate, calcium hydroxide.
- Why are brackets used in $\text{Al}_2(\text{SO}_4)_3$?
- State the valency of: Al, Cl, sulphate, ammonium.